เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 — Year: 2018

Subject: 392131 Physics I — Section: 15 - 18

Date: 1 October 2018 — Time: 10:00-12:00 A.M.

Name:______________________ ID:______________ Field of Study:________

Instructions:

1. The examination has 7 pages (including this page) and a total score of 60 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are NOT allowed to leave the examination room during the first 1 hour after the beginning of the exam.
5. You are NOT allowed to open the exam papers or start to answer before the proctor's permission.
6. You are NOT allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Student who ate caught cheating will also be denied registering for the following semester
11. Express your answer in **English**, other languages will be discarded.
12. If not specified otherwise, used these values: $g = 10\ m/s^2$, $\pi = 3.14$.

Score Report	
1	
2	
3	
4	
5	
6	
Total	

Cheating in the exam is considered an extremely serious Offence which will result in expulsion from the University

(1.1) The traffic light turns green, and the driver of a high-performance car slams the accelerator to the floors. The accelerometer registers 30.0 m/s^2. Convert this reading to km/min^2. (2 points)

(1.2) The speed of light is now defined to be 2.99792458×10^8 m/s. Express the speed of light to

(a) 4 significant figures .... (2 points)

(b) 6 significant figures .... (2 points)

(1.3) The edges of a shoebox are measured to be 11.5 cm, 17.7 cm, and 28 cm. Determine the volume of the box retaining the proper number of significant figures in your answer. (2 points)

(1.4) Use the rules for significant figures to find the answer to the addition problem. (2 points)

$$21.5 + 16 + 117.12 + 4.012 = \ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

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(2.1) I_1 and I_2 are two currents coming into a junction in a circuit. The current I going out of the junction is given by

$$I = I_1 + I_2$$

In an experiment, the values of I_1 and I_2 are determined as 2.0 ± 0.1 A and 1.5 ± 0.2 A respectively. What is the value of I ? What is the uncertainty in this value? **(5 points)**

(2.2) In an experiment, a liquid is heated electrically, causing the temperature to change from 20.0 ± 0.2 °C to 21.5 ± 0.5 °C. Find the change of temperature, with its associated uncertainty. **(5 points)**

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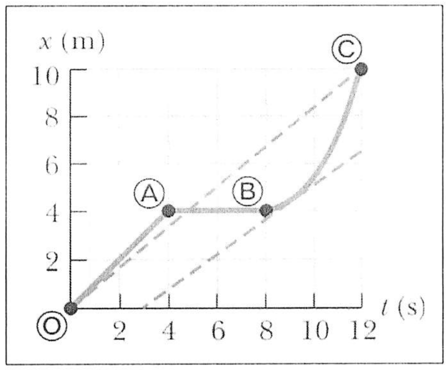
(3) A train moves slowly along a straight portion of track according to the graph of position versus time in figure as shown. **(10 points)**



Find

(a) The average velocity for the total trip.

(b) The average velocity during the first 4 s of motion.

(c) The average velocity during the next 4 s of motion

(d) The instantaneous velocity at t = 2 s

(e) The instantaneous velocity at t = 9 s

(4.1) A hare and a tortoise compete in a race over a straight course 1 km long. The tortoise crawls at a speed of 0.2 m/s toward the finish line. The hare runs at a speed of 8 m/s toward the finish line for 0.8 km and then stops to tease the slow-moving tortoise as the tortoise eventually passes by. The hare waits for a while after the tortoise passes and then runs toward the finish line again at 8 m/s. Both the hare and the tortoise cross the finish line at the exact same instant. Assume both animals, when moving, move steadily at their respective speeds.

(a) How far is the tortoise from the finish line when the hare resumes the race? (2 points)

(b) For how long in time was the hare stationary? (3 points)

(4.2) As soon as a traffic light turns green, a car speeds up from rest to 50 m/s with constant acceleration 5 m/s^2. In the adjoining bicycle lane, a cyclist speeds up from rest to 32 m/s with constant acceleration 8 m/s^2. Each vehicle maintains constant velocity after reaching its cruising speed.

(a) For what time interval is the bicycle ahead of the car? (2 points)

(b) By what maximum distance does the bicycle lead the car? (3 points)

(5.1) A stone is thrown straight downward with initial speed 5.0 m/s from a height of 30.0 m. Find *(a)* the time it takes to reach the ground and *(b)* the speed with which it strikes.

(5 points)

(5.2) A bottle dropped from a balloon reaches the ground in 20 s. Determine the height of the balloon if

(a) It was at rest in the air.

(b) It was ascending with a speed of 50 m/s when the bottle was dropped.

(5 points)

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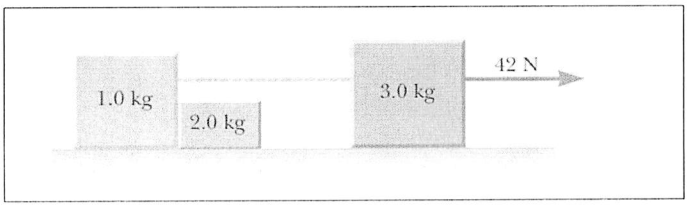
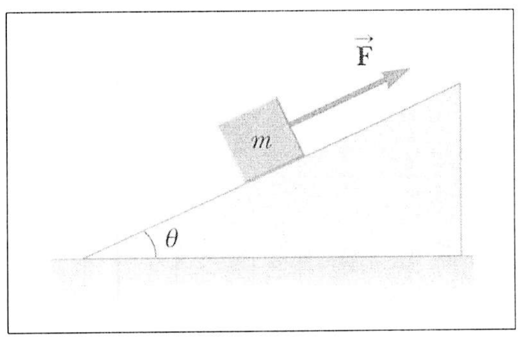
(6.1) Assume the three blocks portrayed in Figure as shown move on a frictionless surface and a 42-N force acts as shown on the 3.0-kg block. Determine (a) the acceleration given this system, (b) the tension in the cord connecting the 3.0-kg and (c) the 1.0-kg blocks and the force exerted by the 1.0-kg block on the 2.0-kg block. (6 points)



(6.2) A block of mass $\boldsymbol{m}$ = 6 kg is pulled up a $\boldsymbol{\theta}$ = 37° incline as shown in figure with a force of magnitude $\boldsymbol{F}$ = 48 N.



(a) Find the acceleration of the block if the incline in frictionless. (2 points)

(b) Find the acceleration of the block if the coefficient of kinetic friction between the block and incline is 0.15 (2 points)